

Ex. No: 05

Implementation of filters

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```
In [ ]: # Import the libraries
import cv2
import numpy as np
import matplotlib.pyplot as plt
```

```
In [3]: # Read the Input Image
image = cv2.imread("Eiffel Tower.jpeg")
```

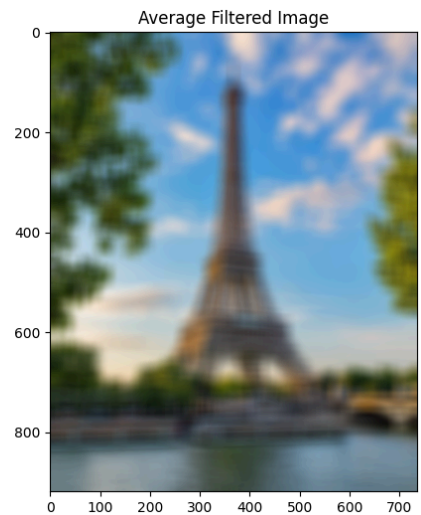
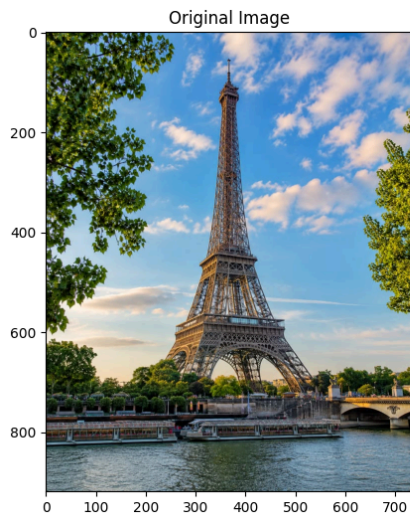
```
In [4]: # Convert the image from BGR to RGB format for display
img1 = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
plt.imshow(img1)
plt.axis('off')
plt.show()
```



```
In [5]: # Average Filter
avg_filter = cv2.blur(img1, (30, 30))
```

```
In [6]: plt.figure(figsize=(18, 6))
plt.subplot(121); plt.imshow(img1); plt.title('Original Image')
plt.subplot(122); plt.imshow(avg_filter); plt.title('Average Filtered Image')
```

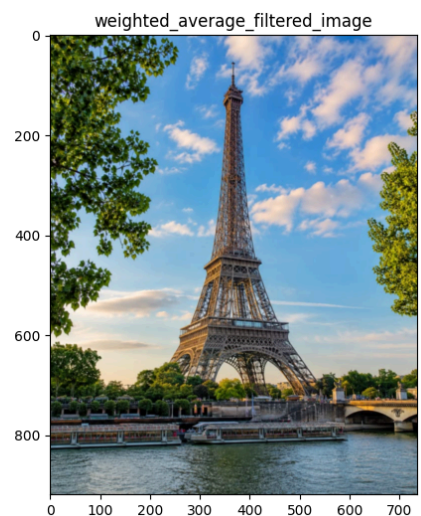
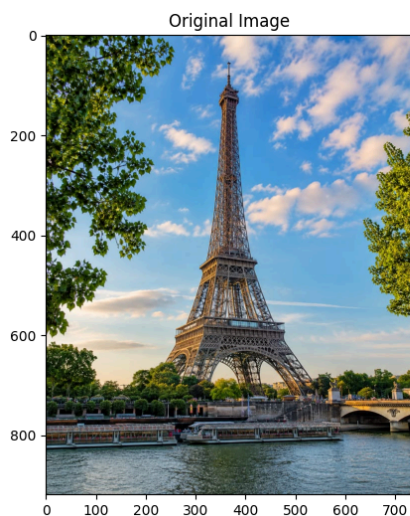
```
Out[6]: Text(0.5, 1.0, 'Average Filtered Image')
```



```
In [7]: # Weighted Average Filter
kernel = np.array([[1, 2, 1],
                   [2, 4, 2],
                   [1, 2, 1]]) / 16

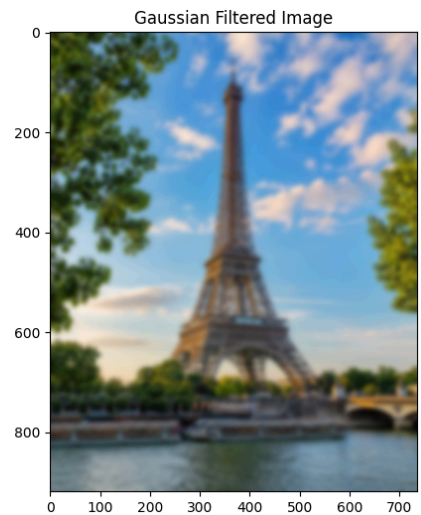
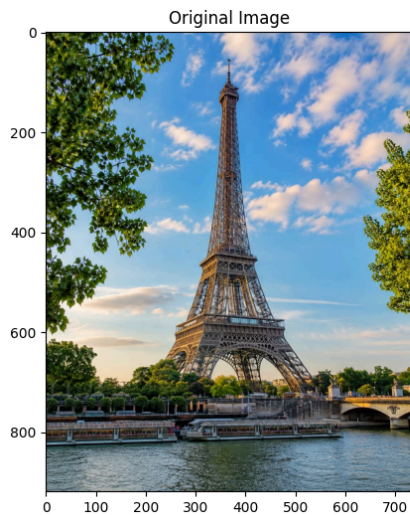
weighted_avg_filter = cv2.filter2D(img1, -1, kernel)
```

```
In [8]: plt.figure(figsize = (18, 6))
plt.subplot(121);plt.imshow(img1); plt.title('Original Image')
plt.subplot(122);plt.imshow(weighted_avg_filter); plt.title('weighted_average_fi
plt.show()
```



```
In [9]: # Gaussian Filter
gaussian_filter = cv2.GaussianBlur(img1, (29, 29), 0, 0)
```

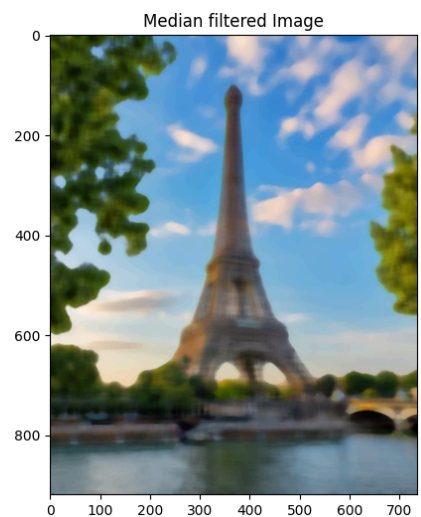
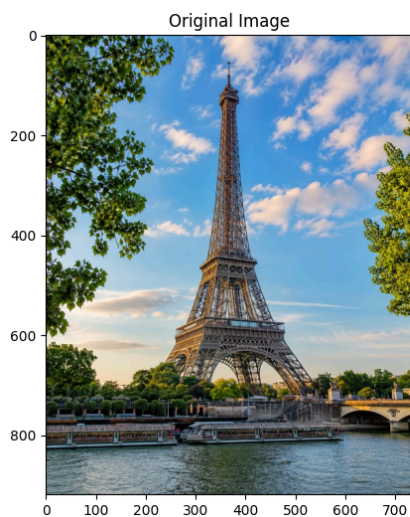
```
In [10]: plt.figure(figsize=(18, 6))
plt.subplot(121); plt.imshow(img1); plt.title('Original Image')
plt.subplot(122); plt.imshow(gaussian_filter); plt.title('Gaussian Filtered Imag
plt.show()
```



```
In [11]: #Median Filter
median_filter = cv2.medianBlur(img1, 19)
```

```
In [12]: plt.figure(figsize = (18, 6))
plt.subplot(121); plt.imshow(img1); plt.title('Original Image')
plt.subplot(122); plt.imshow(median_filter); plt.title('Median filtered Image')
```

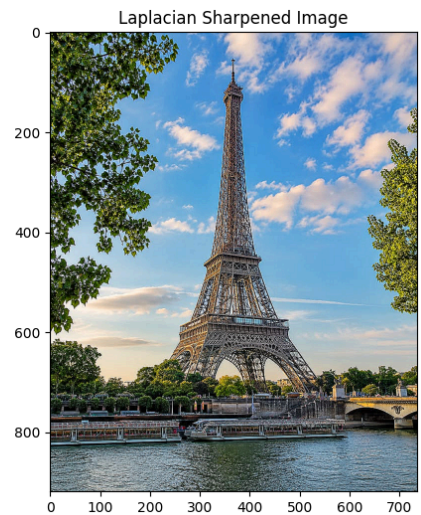
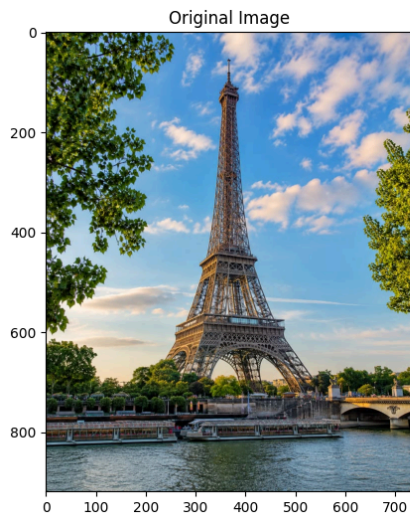
Out[12]: Text(0.5, 1.0, 'Median filtered Image')



```
In [13]: #Laplacian Sharpening
laplacian_kernel = np.array([[0, -1, 0],
                             [-1, 5, -1],
                             [0, -1, 0]])

sharpened_laplacian_kernel = cv2.filter2D(img1, -1, kernel = laplacian_kernel)
```

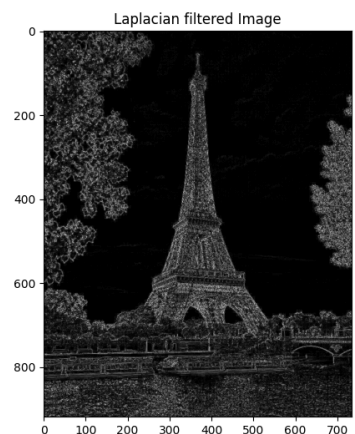
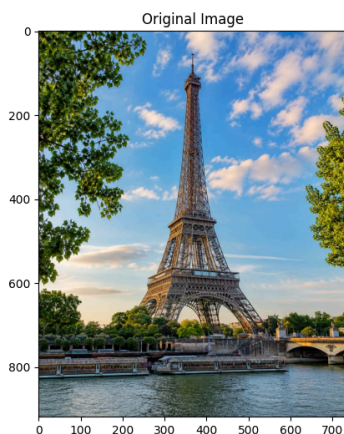
```
In [14]: plt.figure(figsize = (18, 6))
plt.subplot(121); plt.imshow(img1); plt.title('Original Image')
plt.subplot(122); plt.imshow(sharpened_laplacian_kernel); plt.title('Laplacian S
plt.show()
```



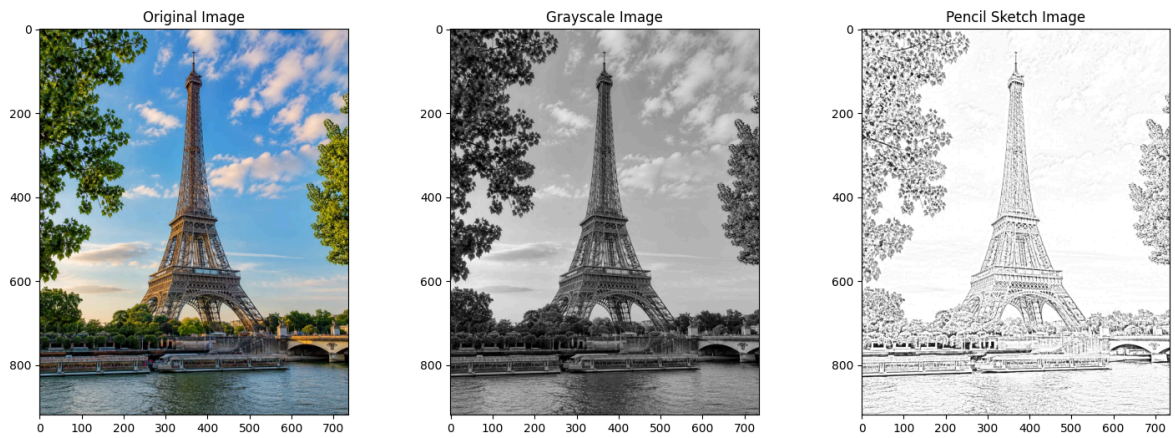
```
In [15]: #Laplacian filter for Grayscale Image
gray_image = cv2.cvtColor(img1, cv2.COLOR_RGB2GRAY)
laplacian_operator = cv2.Laplacian(gray_image, cv2.CV_64F)
laplacian_operator = np.uint8(np.absolute(laplacian_operator))
```

```
In [16]: plt.figure(figsize = (18, 6))
plt.subplot(131); plt.imshow(img1); plt.title('Original Image')
plt.subplot(132); plt.imshow(gray_image, cmap='gray'); plt.title('Grayscale Image')
plt.subplot(133); plt.imshow(laplacian_operator, cmap='gray'); plt.title('Laplacian filtered Image')
```

Out[16]: Text(0.5, 1.0, 'Laplacian filtered Image')



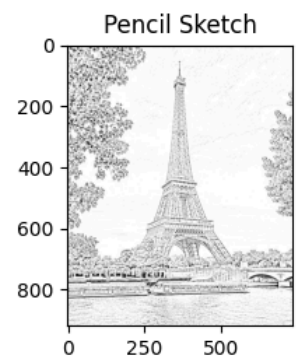
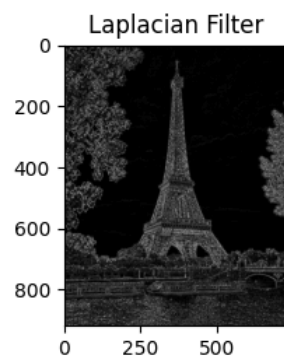
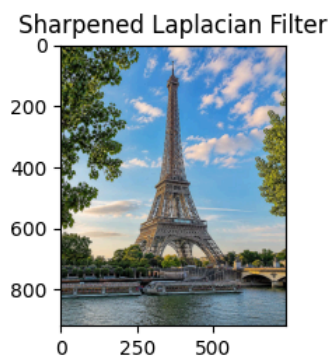
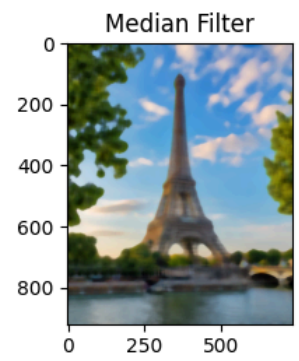
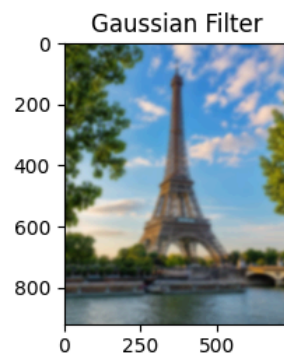
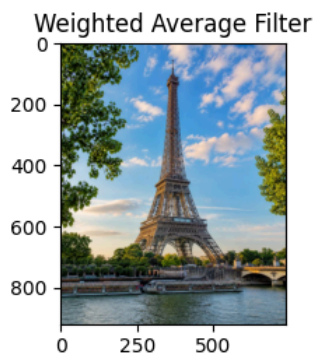
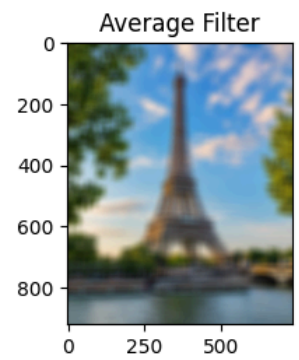
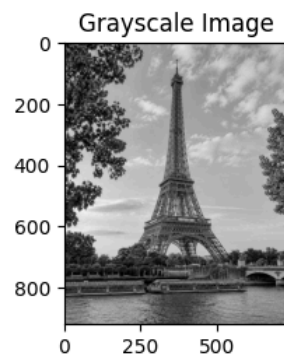
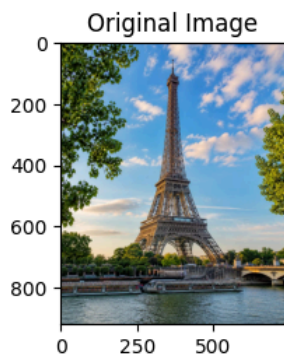
```
In [17]: #Pencil Sketch
# Convert to grayscale
gray = cv2.cvtColor(img1, cv2.COLOR_BGR2GRAY)
# Invert the grayscale image
invert = cv2.bitwise_not(gray)
# Apply Gaussian blur
blur = cv2.GaussianBlur(invert, (21, 21), 0)
# Invert the blurred image
blur = cv2.bitwise_not(blur)
# Create pencil sketch
sketch = cv2.divide(gray, blur, scale=256.0)
# Display images
plt.figure(figsize = (18, 6))
plt.subplot(131); plt.imshow(img1); plt.title('Original Image')
plt.subplot(132); plt.imshow(gray, cmap='gray'); plt.title('Grayscale Image')
plt.subplot(133); plt.imshow(sketch, cmap='gray'); plt.title('Pencil Sketch Image')
plt.show()
```



```
In [18]: #All Images in a grid for a comparison
plt.figure(figsize=(10, 8))

plt.subplot(3,3,1)
plt.imshow(img1); plt.title("Original Image")
plt.subplot(3,3,2)
plt.imshow(gray, cmap="gray"); plt.title("Grayscale Image")
plt.subplot(3,3,3)
plt.imshow(avg_filter); plt.title("Average Filter")
plt.subplot(3,3,4)
plt.imshow(weighted_avg_filter); plt.title("Weighted Average Filter")
plt.subplot(3,3,5)
plt.imshow(gaussian_filter); plt.title("Gaussian Filter")
plt.subplot(3,3,6)
plt.imshow(median_filter); plt.title("Median Filter")
plt.subplot(3,3,7)
plt.imshow(sharpened_laplacian_kernel); plt.title("Sharpened Laplacian Filter")
plt.subplot(3,3,8)
plt.imshow(laplacian_operator, cmap="gray"); plt.title("Laplacian Filter")
plt.subplot(3,3,9)
plt.imshow(sketch, cmap="gray"); plt.title("Pencil Sketch")

plt.tight_layout()
plt.show()
```



In []:

In []: